

Student Number: _____ Class Teacher: _____

St George Girls High School

Trial Higher School Certificate Examination

2017



Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A reference sheet is provided
- In Questions 11 – 15, show relevant mathematical reasoning and/or calculations

Total Marks – 100

Section I Pages 2 – 4

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section
- Answer on the multiple choice answer sheet provided at the back of this paper

Section II Pages 5 – 10

90 marks

- Attempt Questions 11 – 15
- Allow about 2 hours and 45 minutes for this section
- Begin each question in a new writing booklet

Section I	/10
Section II	
Question 11	/18
Question 12	/18
Question 13	/18
Question 14	/18
Question 15	/18
Total	/100

Students are advised that this is a Trial Examination only and does not necessarily reflect the content or format of the Higher School Certificate Examination.

Section I

10 marks

Attempt Questions 1 - 10

Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for Questions 1-10

1. The graph of $y = x(x^2 - 1)$ intersects with the x axis at:
(A) 1 point
(B) 2 points
(C) 3 points
(D) 4 points

2. Which of the following quadratic expressions is positive definite?
(A) $x^2 + 7x + 1$
(B) $x^2 + 7x - 1$
(C) $x^2 + 7x + 15$
(D) $x^2 + 7x - 15$

3. What is the range of the function $f(x) = \sqrt{4 - x^2}$
(A) $0 < y < 2$
(B) $0 \leq y \leq 2$
(C) $-2 < y < 2$
(D) $-2 \leq y \leq 2$

4. The focus of the parabola $x^2 = 8(y - 1)$ is at:
(A) $(0, 1)$
(B) $(0, 3)$
(C) $(0, -1)$
(D) $(0, 8)$

Section I (cont'd)

5. What is the period of $y = \tan 6x$.

- (A) $\frac{\pi}{6}$
- (B) $\frac{\pi}{3}$
- (C) 6π
- (D) 12π

6. What is the value of $\int_{-4}^3 |x + 2| \, dx$

- (A) $\frac{21}{2}$
- (B) $\frac{53}{2}$
- (C) $\frac{3}{2}$
- (D) $\frac{29}{2}$

7. If $y = xe^{2x}$ then $\frac{dy}{dx} =$

- (A) $x e^{2x}$
- (B) $2x e^{2x}$
- (C) $(1 + 2x) e^{2x}$
- (D) $(1 + x) e^{2x}$

8. $|2x + 4| = -x + 4$ when solved has:

- (A) no solution
- (B) 1 solution
- (C) 2 solutions
- (D) 3 solutions

Section I (cont'd)

9. If $f(x) = \frac{3x^4 - x}{x^2}$ then $f'(1) =$

- (A) 5
- (B) 7
- (C) 0
- (D) 2

10. $\sum_{r=1}^5 (-1)^r 2^r =$

- (A) 6
- (B) -62
- (C) 22
- (D) -22

End of Section I

Section II

90 marks

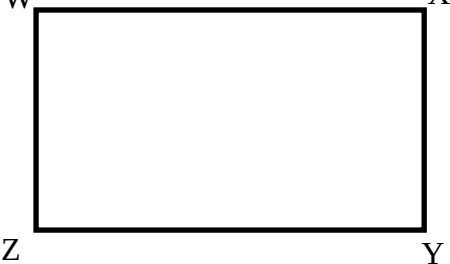
Attempt Questions 11 – 15

Allow about 2 hours and 45 minutes for this section.

Start each question in a new writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

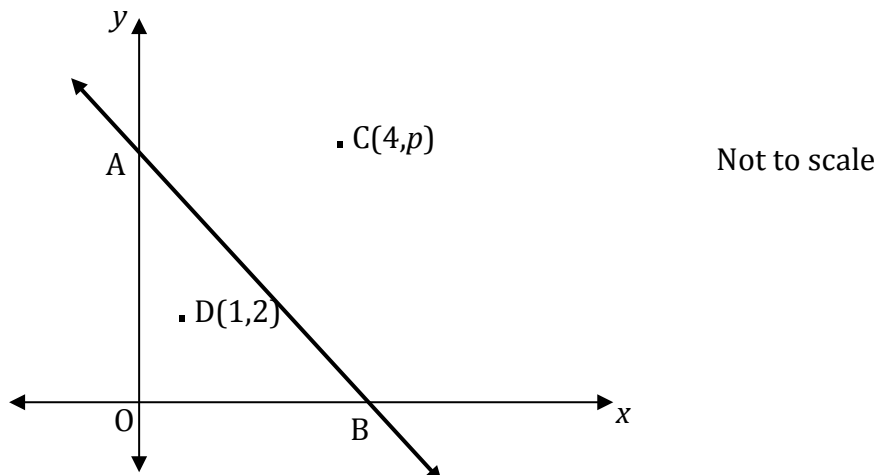
Question 11 (18 marks) Start a New Writing Booklet.	Marks
a) Simplify $\sqrt{75} - \frac{1}{2}\sqrt{48}$.	1
b) Find to 2 decimal places $\sec 40^\circ 15'$.	1
c) Draw a neat sketch of $y = 3 \cos 2x$ for $0 \leq x \leq 2\pi$, showing clearly all relevant features.	2
d) A point $P(x, y)$ moves so that its distance from the x -axis is always twice its distance from the y -axis. Describe this locus geometrically.	1
e) Find the radius of the circle $x^2 + 4x + y^2 - 6y - 12 = 0$.	2
f) Write in simplest form $\frac{x+3}{x^{-1}+3^{-1}}$.	1
g) Differentiate with respect to x	
(i) $\log_e \sqrt{3x^2 - 2}$	(ii) $\frac{x+3}{2x-5}$ 4
h) Prove $\frac{1}{\sin \theta \cos \theta} - \tan \theta = \cot \theta$.	2
i) (i) Find the stationary points of the function $y = 2x^3 - 12x^2 + 18x - 3$ and determine their nature.	3
(ii) In the domain $\{x: -5 \leq x \leq 5\}$ what is the greatest value of $2x^3 - 12x^2 + 18x - 3$?	1

- Question 12** (18 marks) Start a New Writing Booklet. **Marks**
- a) State the domain of x if $x = 3^y$. 1
- b) Find the area between the curve $y = e^x - 2$ and the x -axis from $x = 0$ and $x = 3$. 3
- c) For the arithmetic sequence 400, 350, 300, ... find:
- (i) An expression for T_n . 1
- (ii) Which is the first negative term of the sequence? 2
- (iii) The sum of the first 20 terms. 1
- d) A particle moves along the x axis with acceleration $(t - 2)$ m/s². Initially it is 1 m to the right of the origin, with velocity 3 m/s. What is the position of the particle after 6 seconds? 3
- e)  WXYZ is a rectangle. Prove $XZ = WY$. 3
- f) Solve $x^2 > 3x$. 2
- g) For what values of x is $y = x^3 - 3x + 5$ an increasing function? 2

Question 13 (18 marks) Start a New Writing Booklet.

Marks

a)



The line $2x + 3y = 12$ cuts the x -axis at B and the y -axis at A.

- | | |
|---|---|
| (i) Calculate the length of AB as a simplified surd. | 1 |
| (ii) If AC is perpendicular to AB, find the value of p if C is the point $(4,p)$ and D $(1,2)$. | 2 |
| (iii) Calculate the perpendicular distance from D to AB. | 2 |
| (iv) Hence, or otherwise, find the area of $\triangle ABD$. | 1 |
| (v) Draw this diagram in your answer booklet and shade the region $2x + 3y < 12$. | 1 |
| b) If $0 \leq \theta \leq 2\pi$ solve $\sin 2\theta = -\frac{\sqrt{3}}{2}$. | 2 |
| c) If α and β are the roots of $3x^2 - 4x - 1 = 0$ find the values of: | |
| (i) $\alpha + \beta$ | 1 |
| (ii) $\alpha\beta$ | 1 |
| (iii) $\alpha^2 + \beta^2$ | 1 |
| d) Use Simpson's Rule with 5 function values to approximate giving your answer to 2 significant figures. $\int_1^9 \log_e x \, dx$, | 3 |
| e) A particle moving in a straight line at time t (in seconds) has displacement x (in cm) given by $x = 6t - t^3$. When is the particle at rest and what is the acceleration at that time? | 3 |

Question 14 (18 marks) Start a New Writing Booklet.

Marks

- a) The fourth term of a geometric sequence is 96 and the seventh term is 12.

Find the

- (i) first term and common ratio.

2

- (ii) first term smaller than 0.0001 .

2

- b) Find the equation of the tangents to the curve $y = 4 \cos x$ at the point where $x = \frac{\pi}{6}$

3

- c) Annie was born on the 1st January 2000. Her parents invest \$1000 on this day and on every birthday thereafter. The interest is paid at 6% compounded annually. After completing her HSC she decides to use the account to fund a gap year. She withdraws all the funds on 31/12/17 (getting paid her interest for 2017).

- (i) What is the value of the investment on the 31/12/01
(after the interest for 2001 is paid)?

2

- (ii) How much does Annie collect on 31/12/17?

3

- d) Solve $2 \log_2 x - \log_2(2x + 6) = 1$.

3

- e) Find:

- (i) $\int 2\sin\left(\frac{\pi}{4} + x\right)dx$

1

- (ii) $\int \frac{x}{x^2 + 3} dx$

2

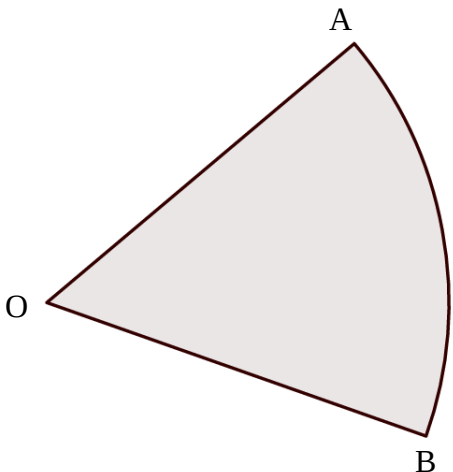
Question 15 (18 marks) Start a New Writing Booklet.

Marks

a) Evaluate $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$ 2

b) For what value of n is $\frac{6^{3n} \times 9^{n+1}}{8^n} = 1$. 2

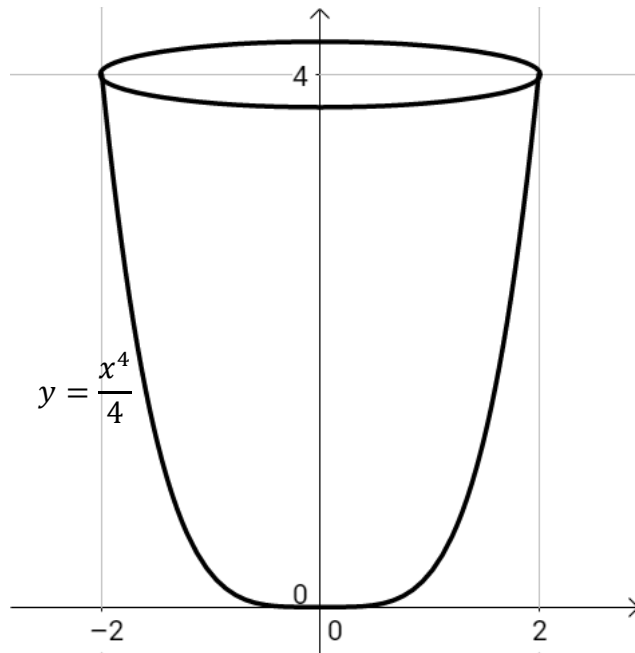
c) The parabola $y = ax^2 + bx + c$ passes through the points $(0, 5)$, $(1, 3)$ and $(-1, 5)$. 3
 Find the value of a, b and c .

d)  2

Arc AB is subtended by an angle of 72° at the centre of a circle radius 8cm.

(i) Calculate the length AB.

(ii) Calculate the area of sector AOB

e)  4

A bowl is formed by rotating the curve $y = \frac{x^4}{4}$ between $x=0$ and $x=2$ about the y axis. Find the volume of the bowl.

- f) A cylindrical container closed at both ends is made from thin sheet metal. The container is to have a radius of r cm and height of h cm, such that its volume is 1000π cm³.

[So $V = \pi r^2 h$ and $SA = 2\pi r^2 + 2\pi r h$]

- (i) Show that the area of sheet metal required to make the container is

$$\left(2\pi r^2 + \frac{2000\pi}{r}\right) \text{ cm}^2 \quad 1$$

- (ii) Hence find the minimum area of sheet metal required to make the container. 4

End of Examination

Mathematics

Factorisation

$$a^2 - b^2 = (a + b)(a - b)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Angle sum of a polygon

$$S = (n - 2) \times 180^\circ$$

Equation of a circle

$$(x - h)^2 + (y - k)^2 = r^2$$

Trigonometric ratios and identities

$$\sin \theta = \frac{\text{opposite side}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent side}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite side}}{\text{adjacent side}}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

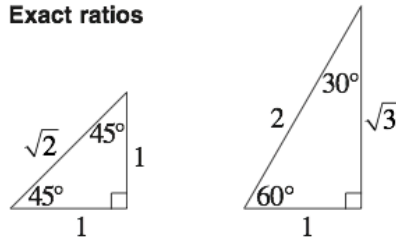
$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

Exact ratios



Sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Cosine rule

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Area of a triangle

$$\text{Area} = \frac{1}{2} ab \sin C$$

Distance between two points

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Perpendicular distance of a point from a line

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Slope (gradient) of a line

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Point-gradient form of the equation of a line

$$y - y_1 = m(x - x_1)$$

n th term of an arithmetic series

$$T_n = a + (n - 1)d$$

Sum to n terms of an arithmetic series

$$S_n = \frac{n}{2} [2a + (n - 1)d] \quad \text{or} \quad S_n = \frac{n}{2} (a + l)$$

n th term of a geometric series

$$T_n = ar^{n-1}$$

Sum to n terms of a geometric series

$$S_n = \frac{a(r^n - 1)}{r - 1} \quad \text{or} \quad S_n = \frac{a(1 - r^n)}{1 - r}$$

Limiting sum of a geometric series

$$S = \frac{a}{1 - r}$$

Compound interest

$$A_n = P \left(1 + \frac{r}{100} \right)^n$$

Mathematics (continued)

Differentiation from first principles

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Derivatives

If $y = x^n$, then $\frac{dy}{dx} = nx^{n-1}$

If $y = uv$, then $\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$

If $y = \frac{u}{v}$, then $\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

If $y = F(u)$, then $\frac{dy}{dx} = F'(u) \frac{du}{dx}$

If $y = e^{f(x)}$, then $\frac{dy}{dx} = f'(x)e^{f(x)}$

If $y = \log_e f(x) = \ln f(x)$, then $\frac{dy}{dx} = \frac{f'(x)}{f(x)}$

If $y = \sin f(x)$, then $\frac{dy}{dx} = f'(x) \cos f(x)$

If $y = \cos f(x)$, then $\frac{dy}{dx} = -f'(x) \sin f(x)$

If $y = \tan f(x)$, then $\frac{dy}{dx} = f'(x) \sec^2 f(x)$

Solution of a quadratic equation

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sum and product of roots of a quadratic equation

$$\alpha + \beta = -\frac{b}{a} \qquad \alpha\beta = \frac{c}{a}$$

Equation of a parabola

$$(x-h)^2 = \pm 4a(y-k)$$

Integrals

$$\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C$$

$$\int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + C$$

$$\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + C$$

$$\int \sin(ax+b) dx = -\frac{1}{a} \cos(ax+b) + C$$

$$\int \cos(ax+b) dx = \frac{1}{a} \sin(ax+b) + C$$

$$\int \sec^2(ax+b) dx = \frac{1}{a} \tan(ax+b) + C$$

Trapezoidal rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{2} [f(a) + f(b)]$$

Simpson's rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

Logarithms – change of base

$$\log_a x = \frac{\log_b x}{\log_b a}$$

Angle measure

$$180^\circ = \pi \text{ radians}$$

Length of an arc

$$l = r\theta$$

Area of a sector

$$\text{Area} = \frac{1}{2} r^2 \theta$$

Student Name: _____ Class Teacher: _____

Section I

Year 12 Trial HSC Examination 2017

Mathematics

Multiple-choice Answer Sheet - Questions 1 – 10

Allow about 15 minutes for this section.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows:

A ☒ B ☒ C ☐ D ☐
 correct ↗

-
- | | | | | | | | | |
|-----|---|-----------------------|---|-----------------------|---|-----------------------|---|-----------------------|
| 1. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 2. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 3. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 4. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 5. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 6. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 7. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 8. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 9. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 10. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |

Student Name: _____ Class Teacher: _____

Section I

Year 12 Trial HSC Examination 2017

Mathematics

Multiple-choice Answer Sheet - Questions 1 – 10

Allow about 15 minutes for this section.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows:

A ☒ B ☒ C ☐ D ☐
 correct

-
- | | | | | |
|-----|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| 1. | A ☐ | B ☐ | C ☒ | D ☐ |
| 2. | A ☐ | B ☐ | C ☒ | D ☐ |
| 3. | A ☐ | B ☒ | C ☐ | D ☐ |
| 4. | A ☐ | B ☒ | C ☐ | D ☐ |
| 5. | A ☒ | B ☐ | C ☐ | D ☐ |
| 6. | A ☐ | B ☐ | C ☐ | D ☒ |
| 7. | A ☐ | B ☐ | C ☒ | D ☐ |
| 8. | A ☐ | B ☐ | C ☒ | D ☐ |
| 9. | A ☐ | B ☒ | C ☐ | D ☐ |
| 10. | A ☐ | B ☐ | C ☐ | D ☒ |

$$1) y = x(x^2 - 1)$$

$$0 = x(x-1)(x+1)$$

$$\therefore x = 0, 1, -1 \quad \therefore 3 \text{ pts}$$

C

$$2) a > 0 \quad b^2 - 4ac < 0$$

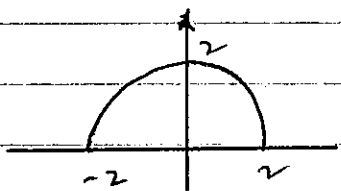
$$7^2 - 4(1)(c) < 0$$

$$49 - 4c < 0$$

$$c > 12\frac{1}{4}$$

C

$$3) y = \sqrt{4 - x^2}$$



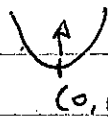
RANGE

B

$$4) x^2 = 4a(y - k)$$

$$4a = 8$$

$$\therefore a = 2$$



$$S(0, 3)$$

B

$$5) y = \tan x$$

$$y = \tan 6x$$

$$D: -\frac{\pi}{2} < x < \frac{\pi}{2}$$

$$R: \text{all real } y$$

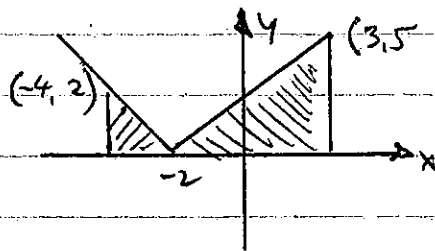
$$-\frac{\pi}{2} < 6x < \frac{\pi}{2}$$

$$-\frac{\pi}{12} < x < \frac{\pi}{12}$$

$$\therefore \frac{\pi}{6}$$

A

$$6) \int_{-4}^3 |x+2| dx$$



$$2 + 12\frac{1}{2} = \frac{29}{2}$$

D

$$7) y = x e^{2x}$$

$$y' = 1 \cdot e^{2x} + 2x e^{2x}$$

$$= e^{2x} (1 + 2x)$$

C

$$8) \quad |2x+4| = -x+4$$

$$2x+4 = -x+4$$

$$3x = 0$$

$$x = 0$$

$$LHS = 4$$

$$RHS = 4$$

✓

$$\text{or } 2x+4 = -(-x+4)$$

$$2x+4 = x-4$$

$$x = -8$$

$$LHS = 12$$

$$RHS = 12$$

✓

∴ 2 solⁿ

C

$$9) \quad f(x) = \frac{3x^4 - x}{x^2}$$

$$= 3x^2 - x^{-1}$$

$$f'(x) = 6x + x^{-2}$$

$$f'(1) = 6 + 1$$

$$= 7$$

B

$$10) \quad \sum_{r=1}^5 (-1)^r 2^r$$

$$= -1(2)^1 + (-1)^2(2)^2 + (-1)^3(2)^3 + (-1)^4(2)^4 + (-1)^5(2)^5$$

$$= -2 + 4 - 8 + 16 - 32$$

$$= 20 - 42$$

$$= -22$$

D

a) $\sqrt{75} - \frac{1}{2}\sqrt{48} = 5\sqrt{3} - \frac{1}{2}(4\sqrt{3})$
 $= 3\sqrt{3}$

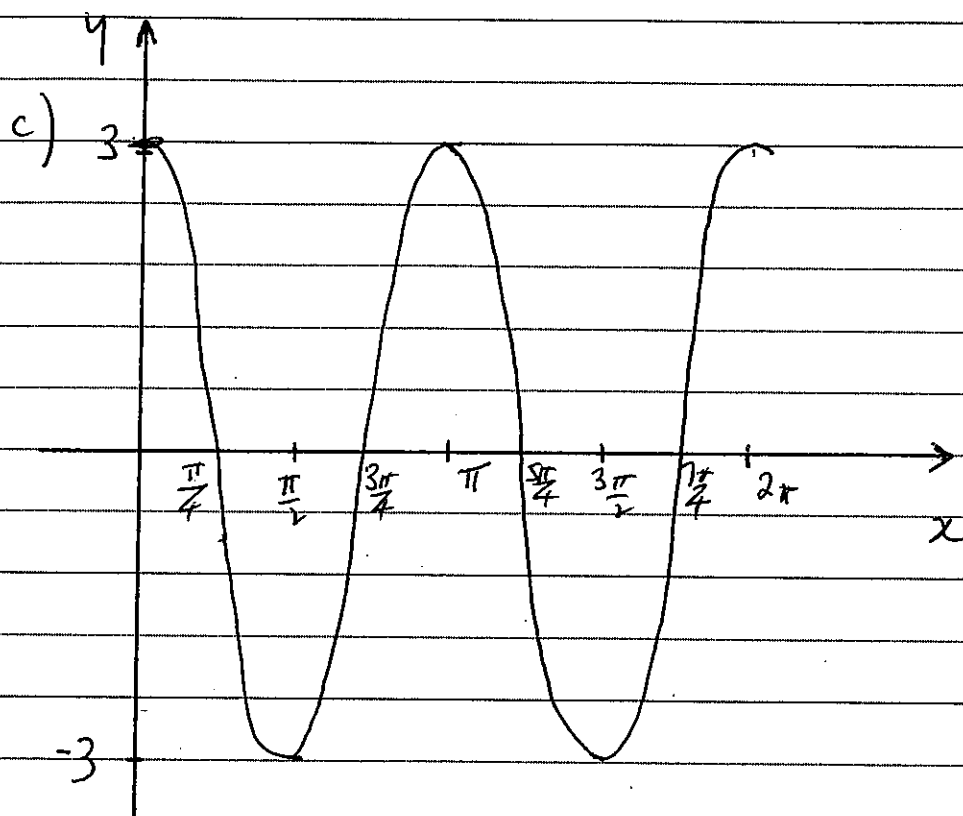
$\frac{1}{2}$

$\frac{1}{2}$

b) $\sec 40^\circ 15' = \frac{1}{\cos 40^\circ 15'}$
 $= 1.31$

1

1.3 $\frac{1}{2}$ MARK



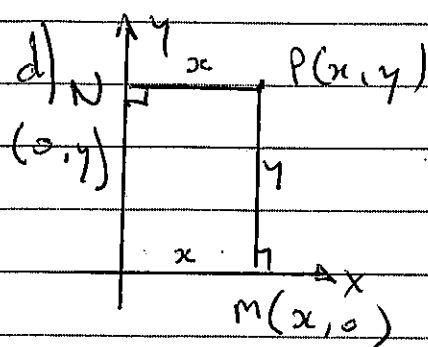
2.

FULL MARKS FOR
CORRECT PERIOD,
AMPLITUDE, SHAPE

$\frac{1}{2}$ $0 \leq \theta \leq \pi$

$\frac{1}{2}$ if 2 or
less x intercepts
shown

NOTE SCALE



$2PN = PM$
 $4PN^2 = PM^2$
 $4x^2 = y^2$

$y = \pm 2x$

or $|y| = |2x|$

$y = \pm 2x$

or $y = |2x|$
 [POORLY DONE]

$\frac{1}{2}$ MARK FOR

$y = 2x$ with
some supporting
working.

locus is two straight lines with
gradients ± 2 passing through (0,0)

MATHEMATICS- QUESTION 11 (cont)

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

$$e) \quad x^2 + 4x + y^2 - 6y - 12 = 0$$

$$x^2 + 4x + 4 + y^2 - 6y + 9 = 12 + 4 + 9$$

$$(x+2)^2 + (y-3)^2 = 25$$

$$\therefore \text{Radius} = 5$$

Mostly well done

$$f) \quad \frac{x+3}{\frac{1}{x} + \frac{1}{3}} \times \frac{3x}{3x} = \frac{3x(x+3)}{3+x}$$

$$= 3x$$

Many did unusual things when trying to take reciprocal

$$g) \quad i) \quad y = \log_e \sqrt{3x^2 - 2}$$

$$\text{Let } m = (3x^2 - 2)^{1/2}$$

$$\frac{dm}{dx} = \frac{1}{2} (3x^2 - 2)^{-1/2} \cdot 6x$$

$$= 3x (3x^2 - 2)^{-1/2}$$

$$y = \log_e m$$

$$\frac{dy}{dm} = \frac{1}{m}$$

$$\frac{dy}{dx} = \frac{dy}{dm} \times \frac{dm}{dx}$$

$$= \frac{1}{m} \times 3x (3x^2 - 2)^{-1/2}$$

$$= \frac{3x}{3x^2 - 2}$$

MATHEMATICS- QUESTION

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

OR

$$y = \log_e \sqrt{3x^2 - 2}$$

$$= \frac{1}{2} \log_e (3x^2 - 2)$$

$$\frac{dy}{dx} = \frac{1}{2} \cdot \frac{1}{3x^2 - 2} \cdot 6x$$

$$= \frac{3x}{3x^2 - 2}$$

2

This is a better method.

ii) $y = \frac{x+3}{2x-5}$

$$u = x+3 \quad v = 2x-5$$

$$u' = 1 \quad v' = 2$$

1

$$\frac{dy}{dx} = \frac{1(2x-5) - 2(x+3)}{(2x-5)^2}$$

1

$$= \frac{2x-5-2x-6}{(2x-5)^2}$$

ERRORS FROM HERE ON ARE IGNORED.

$$= \frac{-11}{(2x-5)^2}$$

ii) $LMS = \frac{1}{\sin \theta \cos \theta} - \tan \theta$

$$= \frac{1}{\sin \theta \cos \theta} - \frac{\sin \theta}{\cos \theta}$$

$$= \frac{1}{\sin \theta \cos \theta} - \frac{\sin^2 \theta}{\sin \theta \cos \theta}$$

$$= \frac{1 - \sin^2 \theta}{\sin \theta \cos \theta}$$

$$= \frac{\cos^2 \theta}{\sin \theta \cos \theta}$$

1

Well done by most

MATHEMATICS- QUESTION 11 cont

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

$$= \frac{\cos \theta}{\sin \theta}$$

$$= \cot \theta$$

$$= R.H.S.$$

1

i) 1) $y = 2x^3 - 12x^2 + 18x + 3$

$$y' = 6x^2 - 24x + 18$$

$$= 6(x^2 - 4x + 3)$$

MUST HAVE 6.

$$= 6(x-3)(x-1)$$

1

$$y' = 0 \quad \text{when } x = 1 \text{ or } 3$$

$$y'' = 12x - 24$$

when $x = 1$ $y'' = -12$

$$< 0 \quad \cap$$

$$\therefore \text{max t.p. } (1, 5)$$

1

when $x = 3$ $y'' = 12$

$$> 0 \quad \cup$$

$$\therefore \text{min t.p. } (3, -3)$$

1

ii) Max value at end points of domain or at local maxima

$$x = 5 \quad y = 2(5)^3 - 12(5)^2 + 18(5) + 3$$

$$= 37$$

$$x = -5 \quad y = 2(-5)^3 - 12(-5)^2 + 18(-5) + 3$$

$$< 0$$

$$\therefore \text{Max value is } 37$$

Many students did not seem to know max/min needs to be checked at end of domain

1

MATHEMATICS— QUESTION 12

SUGGESTED SOLUTIONS

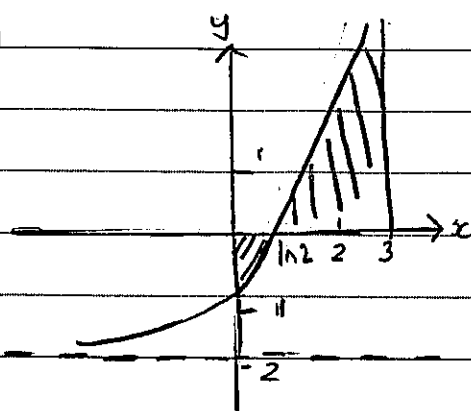
MARKS

MARKER'S COMMENTS

12 a) $x > 0$

1

b)



when $y=0$

$$e^x = 2$$

$$x = \ln 2$$

$$\doteq 0.69 \dots$$

This question was poorly done. Stress to students to always draw a diagram.

$$A = \left| \int_0^{\ln 2} (e^x - 2) dx \right| + \int_{\ln 2}^3 (e^x - 2) dx$$

1

$$= \left| [e^x - 2x]_0^{\ln 2} \right| + [e^x - 2x]_{\ln 2}^3$$

Took off $\frac{1}{2}$ mark for silly errors

$$= \left| e^{\ln 2} - 2\ln 2 - (e^0 - 2(0)) \right| + [e^3 - 6 - (e^{\ln 2} - 2\ln 2)]$$

$$= |2 - 2\ln 2 - 1| + e^3 - 6 - 2 + 2\ln 2$$

$$= |1 - 2\ln 2| + e^3 - 8 + 2\ln 2$$

$$= 2\ln 2 - 1 + e^3 - 8 + 2\ln 2$$

1

$$= 4\ln 2 + e^3 - 9 \text{ units}^2$$

accepted 13.86 u^2 (to 2dp)

Students received $\frac{1}{3}$ if they had the answer $e^3 - 7$ or 13.085... and did not take into account the negative area.

MATHEMATICS- QUESTION

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

$$\begin{aligned} \text{c) (i) } T_n &= a + (n-1)d \\ &= 400 + (n-1)(-50) \\ &= 450 - 50n \\ \text{or } 5(9-n) \end{aligned}$$

1

$$\text{(ii) } T_n < 0$$

$$450 - 50n < 0$$

$$450 < 50n$$

$$9 < n$$

$$n > 9$$

$$\therefore n = 10$$

The 10th term is the first negative term

$$\begin{aligned} T_{10} &= 400 + (10-1)(-50) \\ &= -50 \end{aligned}$$

1

1

$$\text{(iii) } S_n = \frac{n}{2} [2a + (n-1)d]$$

$$= \frac{20}{2} [2(400) + (20-1)(-50)]$$

$$= -1500$$

1

$$\text{d) } a = t - 2$$

$$v = \frac{t^2}{2} - 2t + c \quad t=0, v=3$$

$$3 = \frac{0^2}{2} - 2(0) + c$$

$$c = 3$$

$$\therefore v = \frac{t^2}{2} - 2t + 3$$

1/2

1/2

MATHEMATICS- QUESTION

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

$$x = \frac{t^3}{6} - t^2 + 3t + C \quad t=0, x=1$$

$\frac{1}{2}$

$$1 = \frac{0^3}{6} - 0^2 + 3(0) + C$$

$$C = 1$$

$$\therefore x = \frac{t^3}{6} - t^2 + 3t + 1$$

$\frac{1}{2}$

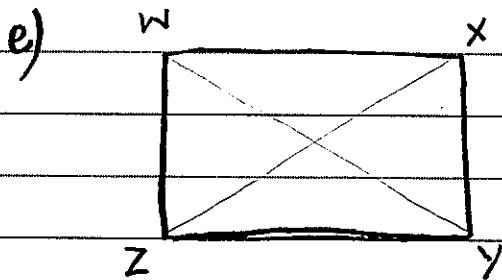
When $t=6$

$$x = \frac{6^3}{6} - 6^2 + 3(6) + 1$$

$$x = 19$$

1

$\therefore$ Position of the particle is 19m to the right of the origin.



Poorly done.

In ΔZWY and ΔZXY

$WZ = XY$ (opposite sides rectangle equal)

ZY is common

$\angle WZY = \angle XYZ = 90^\circ$ (all angles 90° in a rectangle)

$\Delta ZWY \cong \Delta ZXY$ (SAS)

$\therefore XZ = WY$ (corresponding sides in congruent triangles)

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

1

A lot of students chose incorrect triangles to prove congruency. They still achieved marks for knowing congruency proofs and correct final statement.

MATHEMATICS – QUESTION

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

OR

$\angle ZWX = \angle XYZ = 90^\circ$ (all angles 90° in a rectangle)

In ΔWXZ

$$XZ^2 = WZ^2 + WX^2$$

(By Pythagoras)

In ΔWYZ

$$WY^2 = WZ^2 + ZY^2$$

Now $WX = ZY$ (opposite sides rectangle equal)

$$\therefore WY^2 = WZ^2 + WX^2$$

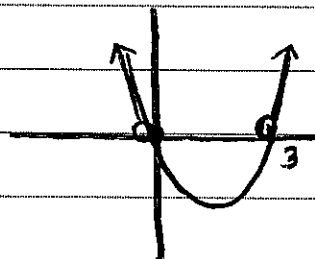
$$WY^2 = XZ^2$$

$$\therefore WY = XZ$$

f) $x^2 - 3x > 0$

$$x(x-3) > 0$$

considering — ①
both solutions.



$$\therefore x > 3 \text{ or } x < 0$$

①/2

①/2

If $x^2 > 3x$

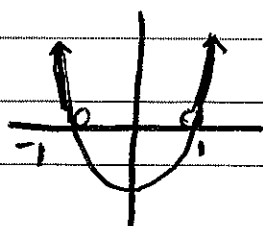
$$x > 3$$

only ①/2

g) $y' = 3x^2 - 3$ increasing $y' > 0$

$$3(x^2 - 1) > 0$$

$$3(x-1)(x+1) > 0$$



$$\therefore x > 1 \text{ or }$$

$$x < -1$$

only ①/2
if equality
sign used.

MATHEMATICS- QUESTION 13 2017 2U TRIAL

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
a). i) $2x + 3y = 12$		Generally well done
$y = 0 \quad x = 6 \quad B(6, 0)$		
$x = 0 \quad y = 4 \quad A(0, 4)$	$\frac{1}{2}$	
$AB^2 = 6^2 + 4^2$		A few students wrote distance formula incorrectly
$= 52$		
$AB = \sqrt{52}$		
$= 2\sqrt{13}$	$\frac{1}{2}$	Surd should be simplified.
ii) $m_{AB} = \frac{0-4}{6-0}$		
$= -\frac{2}{3}$	$\frac{1}{2}$	Use $\frac{y_2 - y_1}{x_2 - x_1}$ or
$m_{DC} = \frac{3}{2}$	$\frac{1}{2}$	not other methods.
$\frac{3}{2} = \frac{p-2}{4-1}$	$\frac{1}{2}$	
$9 = 2p - 4$		Some students did not use $AB \perp DC$ then
$13 = 2p$		$m_{AB} \times m_{DC} = -1$
$p = 6\frac{1}{2}$	$\frac{1}{2}$	
OR $y - 2 = \frac{3}{2}(x - 1)$	$\frac{1}{2} m = \frac{3}{2}$	Usually well done.
$2y - 4 = 3x - 3$		
$0 = 3x - 2y + 1$		
$C(4, p)$		
$0 = 12 - 2p + 1$	1	
$2p = 13$		
$p = 6\frac{1}{2}$	$\frac{1}{2}$	

MATHEMATICS- QUESTION

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

$$iii) d = \frac{|2(1) + 3(2) - 12|}{\sqrt{4+9}}$$

1

Some students did not know how to apply this formula.

$$= \frac{|2+6-12|}{\sqrt{13}}$$

$$= \frac{4\sqrt{13}}{13}$$

1

or $\frac{4}{\sqrt{13}}$

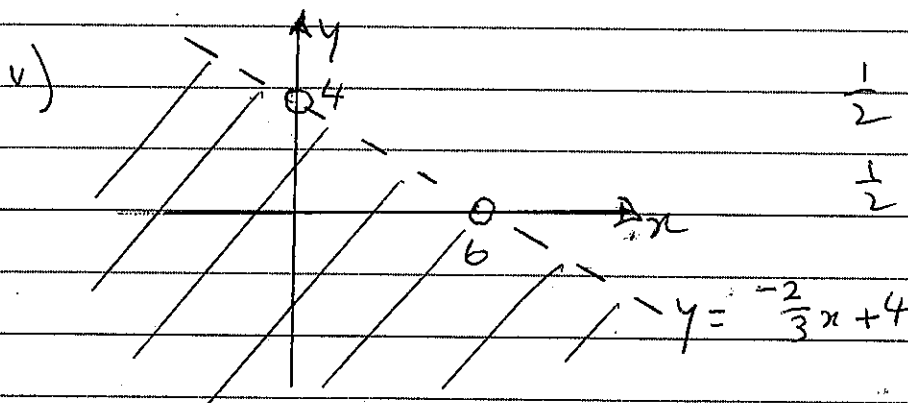
$$iv) A = \frac{1}{2} \times 2\sqrt{13} \times \frac{4}{\sqrt{13}}$$

$\frac{1}{2}$

$$= 4$$

$\frac{1}{2}$

Area is 4 units²



$\frac{1}{2}$ mark dotted line

$\frac{1}{2}$ shading.

$$b) 0 \leq \theta \leq 2\pi$$

$\frac{1}{2}$ mark

Not well done.

$$0 \leq 2\theta \leq 4\pi$$

any correct answer

Majority only had 2 answers

$\sin 2\theta = -\frac{\sqrt{3}}{2}$ Quadrants 3, 4, 7, 8
First quadrant related angle $\frac{\pi}{3}$

1 all

4 values 2θ

$$2\theta = \pi + \frac{\pi}{3}, 2\pi - \frac{\pi}{3}, 3\pi + \frac{\pi}{3}, 4\pi - \frac{\pi}{3}$$

1 4

correct answers.

$$= \frac{4\pi}{3}, \frac{5\pi}{3}, \frac{10\pi}{3}, \frac{11\pi}{3}$$

$$\theta = \frac{2\pi}{3}, \frac{5\pi}{6}, \frac{5\pi}{3}, \frac{11\pi}{6}$$

MATHEMATICS EXTENSION I – QUESTION

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

$$c) i) \alpha + \beta = \frac{4}{3}$$

1

Some errors with signs.

$$ii) \alpha\beta = -\frac{1}{3}$$

1

$$iii) \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$\frac{1}{2}$

$$= \frac{16}{9} + \frac{2}{3}$$

$$= \frac{22}{9}$$

$\frac{1}{2}$

$$d) A \approx \frac{4}{6} (\ln 1 + 4 \ln 3 + \ln 5) + \frac{4}{6} (\ln 5 + 4 \ln 7 + \ln 9)$$

1

Some did not know $\log_e x = \ln x$

$$= \frac{2}{3} (0 + 4 \ln 3 + \ln 5) + \frac{2}{3} (\ln 5 + 4 \ln 7 + \ln 9)$$

$$\approx 11.73$$

1

Some mixed up formula.

$$= 12$$

(signifying)

$$e) x = 6t - t^3$$

$$\dot{x} = 6 - 3t^2$$

1 for differentiation and $\dot{x} = 0$

$$\ddot{x} = 0$$

$$3t^2 = 6$$

1

$$t = \pm\sqrt{2} \quad \text{as } t \geq 0$$

$\frac{1}{2}$

$$t = \sqrt{2}$$

$\frac{1}{2}$

$$\ddot{x} = -6t$$

at

$$t = \sqrt{2}$$

$$\ddot{x} = -6\sqrt{2}$$

$$\text{Acceleration} = -6\sqrt{2} \text{ cm/s}^2$$

1

MATHEMATICS- QUESTION 14

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

$$(a)(i) \text{ GP: } T_4 = ar^3 = 96$$

$$T_7 = ar^6 = 12$$

$$\frac{T_7}{T_4} = \frac{ar^6}{ar^3} = \frac{12}{96}$$

$$r^3 = \frac{1}{8}$$

$$\therefore r = \frac{1}{2} \text{ --- } \textcircled{1} \text{ for correct common ratio}$$

$$a\left(\frac{1}{8}\right) = 96$$

$$a = 96 \times 8$$

$$\therefore a = 768 \text{ --- } \textcircled{1} \text{ for correct first term}$$

$$(ii) ar^{n-1} < 0.0001$$

$$768\left(\frac{1}{2}\right)^{n-1} < 0.0001$$

$$\left(\frac{1}{2}\right)^{n-1} < \frac{0.0001}{768} \text{ --- } \textcircled{1}$$

$$n-1 \left(\ln \frac{1}{2}\right) < \ln\left(\frac{0.0001}{768}\right)$$

$$n-1 > \frac{\ln\left(\frac{0.0001}{768}\right)}{\ln\left(\frac{1}{2}\right)}$$

NOTE: inequality sign reverses as $\ln \frac{1}{2} < 0$.

$$n > \frac{\ln\left(\frac{0.0001}{768}\right)}{\ln\left(\frac{1}{2}\right)} + 1$$

$$n > 23.87267488 \text{ --- } \textcircled{\frac{1}{2}}$$

$$n = 24$$

$$\therefore T_{24} \text{ is the first term}$$

$$< 0.0001$$

--- $\textcircled{\frac{1}{2}}$ for correct answer.

$\left(\frac{1}{2}\right)$ marks if the inequality sign was not reversed, but everything else correct.

MATHEMATICS- QUESTION

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

$$(b) y = 4 \cos x \quad \text{at } x = \frac{\pi}{6} (30^\circ)$$

$$y = 4 \cos \frac{\pi}{6} = \frac{4\sqrt{3}}{2}$$

$$\frac{dy}{dx} = -4 \sin x$$

$$\text{at } x = \frac{\pi}{6},$$

$$y = 2\sqrt{3} \quad \text{--- ① for correct } y \text{ coordinate}$$

$$\frac{dy}{dx} = -4 \sin \frac{\pi}{6}$$

$$= -4 \times \frac{1}{2}$$

$$= -2 \quad \text{--- ① for correct gradient}$$

∴ Equation of tangent:

$$y - 2\sqrt{3} = -2(x - \frac{\pi}{6})$$

$$y - 2\sqrt{3} = -2x + \frac{\pi}{3}$$

$$y = -2x + \frac{\pi}{3} + 2\sqrt{3} \quad \text{--- ① for correct equation}$$

(c)(i) let A_n be the value of the investment after n years.

So A_1 is at 31/12/2000

A_2 is at 31/12/2001

$$A_1 = 1000(1.06) = \$1060 \quad \text{--- ①}$$

$$\begin{aligned} A_2 &= A_1 + A_1(1.06) \\ &= 1000(1.06) + 1000(1.06)^2 \\ &= 1000(1.06)(1 + 1.06) \\ &= \$2183.60 \quad \text{--- ①} \end{aligned}$$

∴ Value on 31/12/2001 is \$2183.60

MATHEMATICS- QUESTION

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

$$(ii) A_1 \rightarrow 31|12|2000$$

$$A_2 \rightarrow 31|12|2001$$

$\vdots$

$$A_{18} \rightarrow 31|12|2017$$

$$\therefore A_{18} = 1000(1.06)^1 + 1000(1.06)^2 + \dots + 1000(1.06)^{18}$$

$$= 1000(1.06) [1 + 1.06 + 1.06^2 + \dots + 1.06^{17}]$$

$$= 1000(1.06) \left[\frac{1(1.06^{18} - 1)}{1.06 - 1} \right] - \textcircled{1}$$

$$= 32\,759.9917$$

$$= \$32\,760.00 - \textcircled{1}$$

①
18

$\rightarrow$ a GP:

$$a = 1$$

$$r = 1.06$$

$$n = 18$$

(2½ marks for A_{17} with correct working).

$$(d) 2 \log_2 x - \log_2 (2x+6) = 1$$

$$\log_2 \left(\frac{x^2}{2x+6} \right) = \log_2 2 - \textcircled{1} \text{ for applying correct log laws.}$$

$$\therefore \frac{x^2}{2x+6} = 2$$

$$x^2 = 4x + 12$$

$$x^2 - 4x - 12 = 0 - \textcircled{1}$$

$$(x+2)(x-6) = 0$$

$$x = -2 \text{ OR } x = 6$$

Test $x = -2$: $2 \log_2 (-2)$ is not defined

$\therefore x = -2$ is not a solution

$$\text{Test } x = 6: 2 \log_2 6 - \log_2 18 = \log_2 \frac{6^2}{18} = \log_2 2 = 1$$

$\therefore x = 6$ is the only solution. - ①

[2½ marks for $x = -2$ and $x = 6$ with no other conclusion]

MATHEMATICS- QUESTION

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

$$(e)(i) \int 2 \sin\left(\frac{\pi}{4} + x\right) dx$$

$$= 2 \times -\cos\left(\frac{\pi}{4} + x\right) + C$$

$$= -2\cos\left(\frac{\pi}{4} + x\right) + C \quad - (1) \text{ for correct answer.}$$

$$(ii) \frac{1}{2} \int \frac{2x}{x^2 + 3} dx \quad - (1) \text{ for showing } \frac{1}{2} \times 2$$

$$= \frac{1}{2} \ln(x^2 + 3) + C \quad - (1) \text{ for correct answer}$$

$$(1) \text{ mark for } \ln(x^2 + 3) + C$$

$$(1) \text{ mark for } 2\ln(x^2 + 3) + C$$

MATHEMATICS – QUESTION 15 (18 marks)

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

pgt.

(a) $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$

2

generally well done
some students did
not set this out
correctly please
improve setting
out in future

$= \lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{(x-2)} \quad (1)$

$= \lim_{x \rightarrow 2} x + 2$

$= 2 + 2$

$= 4 \quad (1)$

(b) $\frac{6^{3n} \times 9^{n+1}}{8^n} = 1$

2

there was a
few different
strategies used.
Very few students
using logarithms
solve the
equation
successfully.

$\frac{(2 \times 3)^{3n} \times (3^2)^{n+1}}{(2^3)^n} = 1 \quad (1)$

$\frac{2^{3n} \cdot 3^{3n} \cdot 3^{2n+2}}{2^{3n}} = 1$

$\frac{2^{3n} \cdot 3^{5n+2}}{2^{3n}} = 1$

$3^{5n+2} = 1$

$3^{5n+2} = 3^0$

$5n + 2 = 0$

$n = -\frac{2}{5} \quad (1)$

Index laws were
not correctly
used.
This topic would
benefit from
revision.

MATHEMATICS – QUESTION 15

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

pg 2.

c) $y = ax^2 + bx + c$ — (1) $(0, 5) (1, 3) (-1, 5)$

3

generally well done

substituting $(0, 5)$ in (1)

$$5 = a(0)^2 + b(0) + c$$

$$\boxed{5 = c} \quad (1)$$

$$\therefore y = ax^2 + bx + 5 \quad (2)$$

substituting, $x = 1$ in (2)
 $y = 3$

$$3 = a + b + 5$$

$$a + b = -2 \quad (3)$$

substituting, $x = -1$ in (2)
 $y = 5$

$$5 = a(-1)^2 + b(-1) + 5$$

$$5 = a - b + 5$$

$$a - b = 0 \quad (4)$$

$$(3) + (4) \quad a + b = -2$$

$$a - b = 0$$

$$2a = -2$$

$$\boxed{a = -1} \quad (1)$$

sub $a = -1$ in (4)

$$a - b = 0$$

$$a = b$$

$$\therefore -1 = b$$

$$\boxed{\therefore b = -1} \quad (1)$$

$$\therefore a = -1, b = -1, c = 5$$

and the equation is

$$y = -x^2 - x + 5$$

MATHEMATICS – QUESTION 15

(a) pg 3 SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
angles in radian measure.		
$360^\circ = 2\pi^c$		• it was very
$1^\circ = \frac{2\pi^c}{360}$		well done by most students
$72^\circ = \frac{2\pi}{360} \times 72^c$		However, a significant
$= \frac{144\pi}{360}$		number of students, incorrectly tried to use degrees
$= \frac{2\pi}{15}$		in the formulas
		$l = r\theta$ and $A = \frac{1}{2}r^2\theta$.
(i) $l = r\theta$ $= 8 \times \frac{2\pi}{5}$	(1)	
$= \frac{16\pi}{5} \text{ cm.}$		
(ii) $A_{\text{sector}} = \frac{1}{2}r^2\theta$	(1)	
$= \frac{1}{2} \times 8^2 \times \frac{2\pi}{5}$		
$= \frac{64\pi}{5} \text{ cm}^2$		
<u>Method 2 (as in junior school)</u>		
$l_{\text{arc}} = \frac{\theta}{360} \times 2\pi r$		
$= \frac{72}{360} \times 2 \times \pi \times 8$		
$= \frac{16\pi}{5} \text{ cm}$		
$A_{\text{sector}} = \frac{\theta}{360} \times \pi r^2$		
$= \frac{72}{360} \times \pi \times 8^2$		
$= \frac{64\pi}{5} \text{ cm}^2$		

MATHEMATICS – QUESTION 15 (18 marks)

pg #	SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS	
e)	$y = \frac{x^4}{4}$ $4y = x^4$ $\therefore x^2 = \sqrt{4y}$ $x^2 = 2\sqrt{y}$ $x = 2y^{\frac{1}{2}}$	$y = \frac{z^4}{4}$ when $z=0$, $y=0$ $x = 2y^{\frac{1}{2}} = \frac{2^4}{4}$ $= 4$	4.	
	$V = \pi \int_0^4 y^2 dx$ $= \pi \int_0^4 2y^{\frac{1}{2}} dx$ $= 2\pi \int_0^4 y^{\frac{1}{2}} dx$ $= 2\pi \left[\frac{2y^{\frac{3}{2}}}{3} \right]_0^4$ $= \frac{4\pi}{3} \left[y^{\frac{3}{2}} \right]_0^4$ $= \frac{4\pi}{3} \left[4^{\frac{3}{2}} - 0^{\frac{3}{2}} \right]$ $= \frac{4\pi}{3} \left[2^3 - 0 \right]$ $= \frac{32\pi}{3}$	integrating correctly (1) evaluating integral. (1)		
f)	$V = \pi r^2 h$ $\frac{1000\pi}{\pi r^2} = \frac{\pi r^2 h}{\pi r^2}$ $\therefore h = \frac{1000}{r^2}$	$S.A = 2\pi r^2 + 2\pi r h$ $= 2\pi r^2 + 2\pi r \cdot \frac{1000}{r^2}$ $= 2\pi r^2 + \frac{2000\pi}{r}$	(1)	Show all necessary steps to receive full marks.

MATHEMATICS – QUESTION 15

pg5. SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
f (ii). $A = 2\pi r^2 + \frac{2000\pi}{r}$	(4)	generally well done.
$= 2\pi r^2 + 2000\pi r^{-1}$		Most problems for students occurred
$\frac{dA}{dr} = 4\pi r - 2000\pi r^{-2}$ (1)		when they were working with
if $\frac{dA}{dr} = 0$,		<u>indices</u> .
$4\pi r - 2000\pi r^{-2} = 0$		setting out
$4\pi r = \frac{2000\pi}{r^2}$		could also be better. Ask
$\frac{4\pi r^3}{4\pi} = \frac{2000\pi}{4\pi}$		for another booklet, even if
$r^3 = 500$		it is your last question,
$r = \sqrt[3]{500}$ (1)		rather than squashing
$\frac{d^2A}{dr^2} = 4\pi + 4000\pi r^{-3}$		you work and perhaps
When $r = \sqrt[3]{500}$		causing mistakes.
$\frac{d^2A}{dr^2} = 4\pi + 4000\pi [500]^{-\frac{1}{3}}$		
$= 4\pi + 4000\pi [500^{\frac{1}{3} \times -3}]$		
$= 4\pi + 4000\pi [500^{-1}]$		
$= 4\pi + \frac{4000\pi}{500}$		
$= 12\pi$ which is positive (1)		
$\therefore$ when $r = \sqrt[3]{500}$ there is a minimum		
where $r = \sqrt[3]{500}$ $A = 2\pi r^2 + \frac{2000\pi}{r}$		
$= 2\pi(500)^{\frac{2}{3}} + 2000\pi(500)^{-\frac{1}{3}}$		
$= \frac{3000\pi}{\sqrt[3]{500}} \text{ cm}^3 = 1187 \text{ cm}^3$		
(nearest cm^3)		